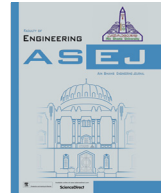




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Sunflower optimization algorithm-based optimal PI control for enhancing the performance of an autonomous operation of a microgrid



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ABSTRACT

This paper presents a novel application of the sunflower optimization (SFO) algorithm to enhance the performance of the inverter based microgrid. The vector cascaded control method is used to control the inverter which relies on the proportional-integral (PI) controller. The main goal is to select the parameters of the PI controller using the SFO algorithm. The multi-objective function for this research is deduced from the response surface methodology (RSM). The simulation results are tested under three different operating states which are: 1) the system conversion from grid-connected to stand-alone mode, 2) changing the load during stand-alone mode, and 3) symmetrical fault during stand-alone mode. The results verify the flexibility, justification, and applicability of the presented SFO algorithm versus the particle swarm optimization (PSO).

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1. Introduction

With the growth of the demand, the large centralized generating stations keep struggling to catch that growth in demands, so the generation trend now is looking forward to the smaller distributed generation (DG) systems which are located near to the load centers. This reduces the transmission losses and decreases the need to expand the transmission network. The DG systems importance increased due to their help in enhancing the reliability, power quality, voltage profile, economic and environmental profits. Mainly, the DG system is capable of:

- 1) Improving the environment by reduce the carbon emissions [1–6].
- 2) Offering convenient solutions to meet the load growth.
- 3) Decreasing the expansion in the transmission system which consequently decreases the transmission and distribution losses.

Additionally, if the DG units are fine controlled, this will improve the reliability and power quality of the load [7,8].

Due to the growth of DG penetration and the shining of multiple DG units, the concept of microgrid is brightening up. The microgrid consists of several DG units and local loads. It can be either grid connected or standalone (islanded) [9–16]. In the grid connected mode, the microgrid is connected to the grid. Moreover, the power balance between the load and generation within the microgrid isn't necessary since any power supply shortage will be supplied from the grid and any overflow in the power will be absorbed by the grid. Meanwhile, in the standalone mode, the microgrid is isolated from the utility grid. In this mode, the DG controllers have to maintain the generation - demand balance and should also maintain the voltage and frequency within their boundaries.

In the autonomous mode, advanced control strategies are used to achieve robust and secure operation. Those strategies can be divided into three main groups: (1) droop-based, (2) centralized control, and (3) multivariable and servomechanism techniques. The droop-based one is utilized to manage the flow of the current of the DG units in an islanded microgrid [17–20]. Droop control for DC system has enhanced reliability compared to other control types [21]. The pros of using droop-based control are its capability to control distributed energy resource (DER) units individually without creating communication between them. However, its main disadvantage is inability to handle nonlinear problems such as load dynamics which affects the system transient response [22–25]. On the other hand, the centralized control techniques require existence of high bandwidth communication links and any failure in these

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links can lead to microgrid collapse [26,27]. Lastly, multivariable and servomechanism controllers are used in DG units for standalone mode [28,29]. However, its disadvantage is the high complexity. Meanwhile, in the nonlinear system, the proportional–integral (PI) controller is always first and pioneer among others because of its large stability margin. On the other hand, the PI controller faces great challenge due to the sensitive effect of the parameters tuning. Therefore, to solve this problem, several meta-heuristic algorithms have brightened up such as Tabu search [30], biogeography-based optimization [31,32], gravitational search algorithm (GSA) [33], harmony search algorithm (HSA) [34], grey wolf optimization and differential evolution (DE) [35], genetic algorithm (GA) [36–38], the non-dominated classification genetic algorithm-II (NSGA-II) [39], Taguchi approach [40], Taguchi combined genetic algorithm (GA) [41], teaching–learning based optimization [42], shuffle frog leaping [43], particle swarm optimization (PSO) [44,45], tree seed algorithm (TSA) [46], artificial bee colony (ABC) [47], and salp swarm algorithm [48]. Each of those techniques has its own advantages and disadvantages [49–51].

In this article, the SFO-based optimal PI controller is introduced though conventional cascaded control system to improve the performance of an autonomous operation of a microgrid with several DG units and their loads. The suggested SFO algorithm is chosen to control the parameters of PI controller. The RSM is utilized to

formulate the fitness function of the optimization. The simulation results are tested under three different operating states. These states are: 1) system change from the grid-connected to the stand-alone, 2) changing the load during the stand-alone, and 3) symmetrical fault during stand-alone mode. The results verify the flexibility, justification, and applicability of the presented SFO algorithm versus the particle swarm optimization (PSO). The SFO algorithm code is constructed by the MATLAB software. The validity of the introduced system is proved based on simulation results which are achieved using PSCAD/EMTDC under several operating scenarios of the microgrid.

2. System model

Fig. 1 represents the single line diagram of a microgrid which contains 3 DGs connected with each other through transmission lines. All the DGs are connected to the PCC which is connected to voltage Source (utility grid) through impedance. Each DG contains DC voltage Source connected to pulse width modulation (two level). The PWM is connected to a delta – star connected transformer through a series filter to improve the power quality. The transformer is connected to a three-phase parallel RLC load which is used to represent the local load. The microgrid data is presented in Table 1.

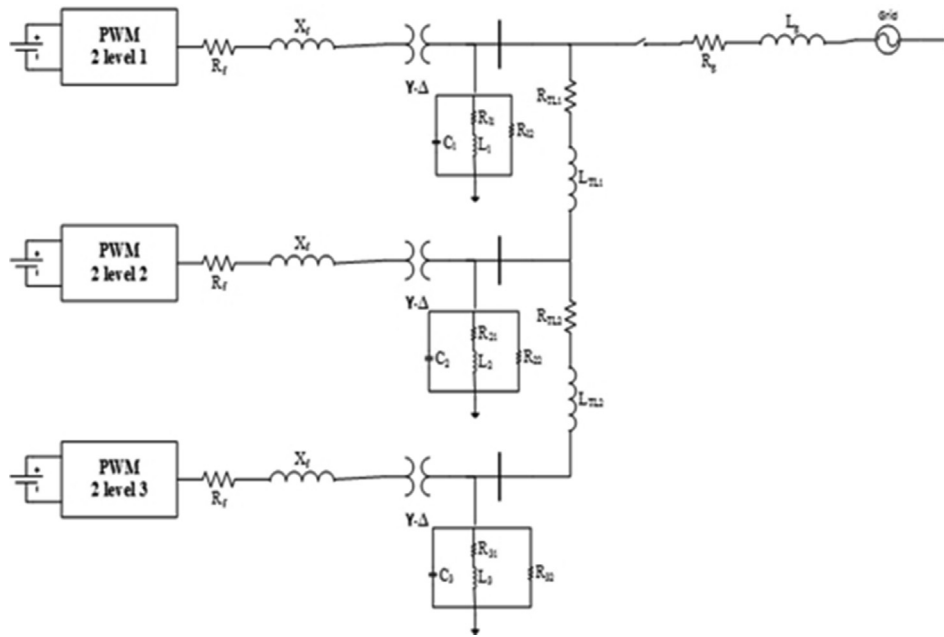


Fig.1. Single line diagram DG system.

Table 1
System data for DG.

$S_{b1} = 2.5$ MVA, $S_{b2} = 5$ MVA, $S_{b3} = 7.5$ MVA, $V_{base\ low} = 0.6$ KV, $V_{base\ high} = 13.8$ KV, $F = 60$ HZ

Transformer parameters

$\Delta/Y = 0.6/13.8$ KV

Load data

Load 1: $C_1 = 50$ μ F, $R_{11} = 9$ Ω , $R_{12} = 150$ Ω , $L_1 = 0.6$ H.

Load 2: $C_2 = 42$ μ F, $R_{22} = 5$ Ω , $R_{12} = 75$ Ω , $L_2 = 0.4$ H.

Load 3: $C_3 = 33$ μ F, $R_{33} = 20$ Ω , $R_{12} = 50$ Ω , $L_3 = 1.5$ H.

Transmission Line data

TL1: $R_{TL1} = 0.7$ Ω , $L_{TL1} = 0.5$ mH.

TL2: $R_{TL2} = 1.5$ Ω , $L_{TL2} = 0.9$ mH.

Filter parameters

$R_f = 1.5$ m Ω , $X_f = 0.5$ m H

Grid data

$V = 13.8$ KV, $R_g = 0.2$ Ω , $L_g = 0.3$ mH

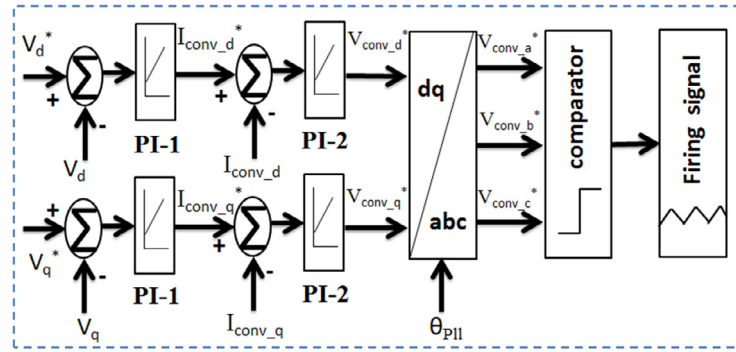


Fig. 2. The conventional control technique for grid-connected and stand-alone modes.

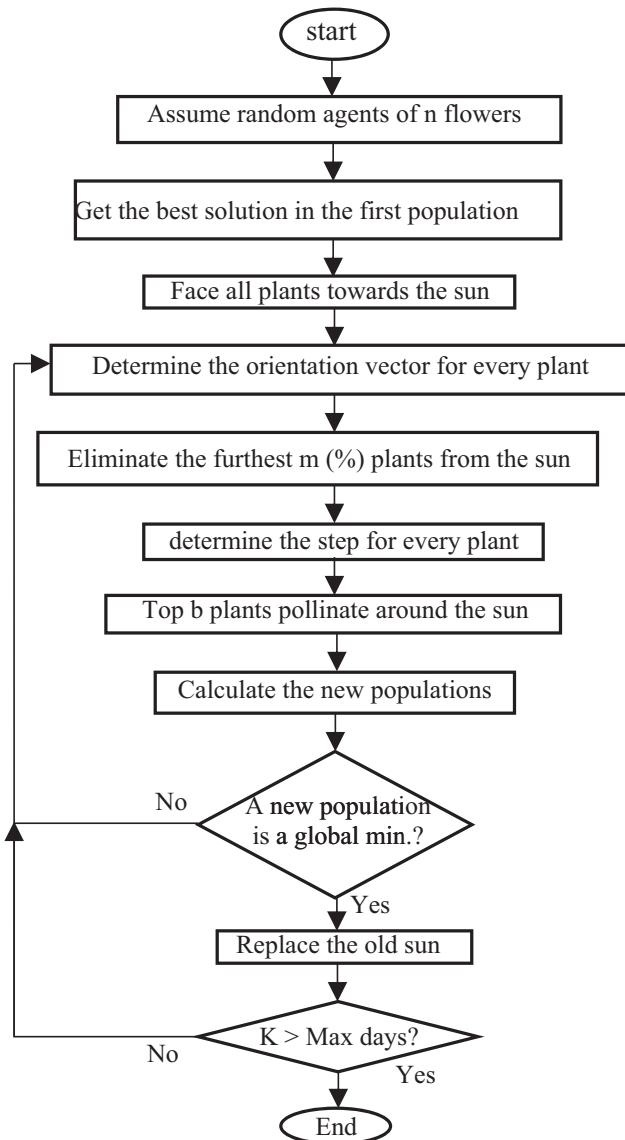


Fig. 3. SFO Algorithm.

3. Control strategy

The main reason of DG instability is the disconnection from the utility grid, which leads to a variation in the ac voltage magnitude and frequency. The conventional cascaded control scheme for each DG in the microgrid is presented in the blue dotted box of Fig. 2. The direct current (I_{conv_d}) maintains the active power, while the quadrature current (I_{conv_q}) controls the terminal voltage of each DG. The reference voltages (V_a, V_b, V_c) are obtained from the transformation between the dq frame and the stationary frame, the transformation angle (θ_{PLL}) is extracted from the phase-locked loop (PLL), which uses the grid-side voltages at its input. These reference voltages are the input of the comparator with 1980 Hz triangular signal to generate the firing pulses of the inverter switches. Fig. 2 shows the conventional control technique for the grid connected and stand-alone modes [40], [52] and [53]. P_{conv}^* is the converter real power reference, P_{conv} is the converter actual real power, I_{conv_d} is the inverter direct axis current, I_{conv_q} is the inverter quadrature axis current, V_d is the direct axis voltage obtained from the transformation of the actual terminal voltage, V_q is the quadrature axis voltage obtained from the transformation of the actual terminal voltage, $V_{conv_d}^*$ is the reference direct axis voltage, and $V_{conv_q}^*$ is the reference of quadrature axis voltage.

4. Modeling stage

4.1. The RSM

The importance of the RSM strategy is rapidly increased due to its success of analyzing different systems. The RSM is considered as a statistical technique. It constructs models empirically by extracting a correlation between the control parameters and their response through a statistical decent technique [54–57]. Through the research study, the PSCAD software helped much in the simulations. It is used to run the RSM model for the scenarios. In this study, the RSM input data are the maximum percentage overshoots (MPOS), maximum percentage undershoots (MPUS), steady-state error (E_{ss}) and settling time (T_{set}) of the terminal voltage profile. For accurate results, the second-order RSM model is used. The central composite design (CCD) technique is used to fit the second-order RSM model [57]. The MINITAB package [58] is used to run the RSM model.

Table 2
RSM designed variables and levels.

Design variable level	K ₁	K ₂	K ₃	K ₄	K ₅	K ₆	K ₇	K ₈	K ₉	K ₁₀	K ₁₁	K ₁₂
Level 1-----(-1)	1.9	0.005	1.5	0.12	1.4	0.004	1.2	0.1	0.9	0.003	0.9	0.09
Level 2-----(-0)	4.2	0.0125	2.25	0.235	3.7	0.0095	1.85	0.2	3.2	0.0065	1.5	0.17
Level 3-----(-1)	6.5	0.02	3	0.35	6	0.015	2.5	0.3	5.5	0.01	2.1	0.25

Table 3

Variables design range and experimental results for Scenario 1 (switching the system from grid connected to stand alone mode of operation).

Exp.	K _{p1}	T _{i1}	K _{p2}	T _{i2}	X ₁ (MPUS ₁)	X ₃ (T _{set1})	X ₄ (E _{ss1})	X ₅ (MPUS ₂)	X ₇ (T _{set2})	X ₈ (E _{ss2})	X ₉ (MPUS ₃)	X ₁₁ (T _{set3})	X ₁₂ (E _{ss3})
1	0	0	0	0	14.87	0.14	0.7517	15.263	0.1485	0.6	15.827	0.15989	0.6278
2	-1	1	1	1	26.51	0.182	1.052	26.8	0.1847	0.732	27.3	0.2202	1.02
3	1	1	1	1	12.23	0.28444	0.693	12.51	0.2984	0.51	12.92	0.3456	0.538
4	-1	-1	-1	-1	18.73	0.1057	0.283	19.1	0.10024	0.37	19.62	0.0864	1.044
5	1	0	0	0	11.572	0.16789	0.8	11.916	0.17872	0.539	12.38	0.19873	0.72086
6	-1	-1	1	1	30.22	0.2931	1.24	30.58	0.3263	0.955	30.96	0.376	0.387
7	0	0	0	0	14.87	0.14	0.7517	15.263	0.1485	0.6	15.827	0.15989	0.6278
8	1	-1	-1	-1	11.41	0.0789	0.377	11.72	0.0796	0.36	12.07	0.084	0.93
9	1	-1	1	1	14.32	0.2624	0.586	14.62	0.2789	0.416	15.091	0.3093	0.658
10	0	0	0	1	14.94	0.14888	0.778	15.288	0.151	0.594	15.803	0.1681	0.448
11	0	0	0	-1	15.38	0.1625	0.483	15.73	0.15156	0.3	16.29	0.1652	0.9867
12	-1	1	1	-1	27.42	0.1817	0.514	27.72	0.18484	0.79	28.23	0.2153	1.54
13	0	0	0	0	14.87	0.14	0.7517	15.263	0.1485	0.6	15.827	0.15989	0.6278
14	0	0	0	0	14.87	0.14	0.7517	15.263	0.1485	0.6	15.827	0.15989	0.6278
15	1	1	-1	-1	9.34	0.0792	0.307	9.68	0.082	0.365	10.1	0.09289	1.12
16	0	0	0	0	14.87	0.14	0.7517	15.263	0.1485	0.6	15.827	0.15989	0.6278
17	1	1	-1	1	8.99	0.0767	0.565	9.314	0.0794	0.3858	9.666	0.08979	0.701
18	-1	-1	-1	1	18.97	0.106	0.664	19.4	0.1055	0.475	19.81	0.0918	0.622
19	-1	0	0	0	23.6	0.1129	0.7733	23.96	0.1155	0.317	24.54	0.1291	0.862
20	1	1	1	-1	12.506	0.2844	0.185	12.79	0.3038	0.738	13.2	0.3537	1.469
21	-1	1	-1	1	13.37	0.0514	0.6	13.7	0.0519	0.4	14.1	0.0564	0.725
22	0	0	0	0	14.87	0.14	0.7517	15.263	0.1485	0.6	15.827	0.15989	0.6278
23	0	-1	0	0	13.68	0.1348	0.8	16.76	0.14	0.584	17.34	0.1483	0.546
24	-1	-1	1	-1	30.31	0.165	0.24	30.63	0.1954	0.51	31.08	0.2458	1.159
25	1	-1	-1	1	11.06	0.0791	1.69	11.37	0.9799	1.31	11.71	0.085	0.73
26	1	-1	1	-1	14.25	0.2789	0.434	14.536	0.29	0.73	14.96	0.3483	1.47
27	0	1	0	0	14.9	0.1457	0.71	15.25	0.15383	0.4566	15.856	0.1704	0.734
28	0	0	0	0	14.87	0.14	0.7517	15.263	0.1485	0.6	15.827	0.15989	0.6278
29	-1	1	-1	-1	12.58	0.0596	0.204	12.89	0.0572	0.526	13.25	0.0622	1.255
30	0	0	1	0	17.2	0.2372	0.719	17.545	0.24578	0.526	18.03	0.2821	0.57089
31	0	0	-1	0	10.61	0.0707	0.5522	10.91	0.0731	0.309	11.305	0.0821	0.8378

Table 4

Variables design range and experimental results for Scenario 2 (variable load mode).

Exp.	K _{p1}	T _{i1}	K _{p2}	T _{i2}	X ₁ (MPUS ₁)	X ₂ (MPOS ₁)	X ₃ (T _{set1})	X ₄ (E _{ss1})	X ₅ (MPUS ₂)	X ₆ (MPOS ₂)	X ₇ (T _{set2})	X ₈ (E _{ss2})	X ₉ (MPUS ₃)	X ₃ (MPOS ₃)	X ₁₁ (T _{set3})	X ₁₂ (E _{ss3})
1	0	0	0	0	2.961	3.955	0.4782	0.266	3.22	3.516	0.49579	0.568	3.97	2.7	0.5872	1.48
2	-1	1	1	1	5.3	7.76	0.6457	0.422	5.33	7.197	0.6567	0.673	5.79	6.31	0.7318	1.3
3	1	1	1	1	2.37	4.16	0.4872	0.55	2.52	3.788	0.5069	0.8005	3.13	3.02	0.6178	1.49
4	-1	-1	-1	-1	4.34	5.143	0.473	0.284	4.52	4.683	0.4821	0.416	5.234	3.8	0.50995	1.47
5	1	0	0	0	2.272	3.323	0.4623	0.207	2.518	2.8879	0.4875	0.639	3.282	2.119	0.59	1.48
6	-1	-1	1	1	5.65	7.67	0.6208	0.324	5.65	7.16	0.6292	0.105	5.99	6.36	0.6737	0.82
7	0	0	0	0	2.961	3.955	0.4782	0.266	3.22	3.516	0.49579	0.568	3.97	2.7	0.5872	1.48
8	1	-1	-1	-1	3.12	3.67	0.473	0.461	3.36	3.19	0.482	0.39	4	2.4	0.5124	1.311
9	1	-1	1	1	2.44	4.85	0.5067	0.178	2.59	4.5	0.51233	0.572	3.13	3.756	0.60666	1.173
10	0	0	0	1	2.8	4.07	0.468	0.437	3.009	3.64	0.4816	0.389	3.63	2.93	0.5066	1.11
11	0	0	0	-1	3.728	4.37	0.5371	1.445	4.08	3.87	0.5597	1.742	5.132	2.8	1	2.931
12	-1	1	1	-1	6.34	7.11	0.729	0.98	6.52	6.6	0.79	1.49	7.36	5.56	1.1	0.45
13	0	0	0	0	2.961	3.955	0.4782	0.266	3.22	3.516	0.49579	0.568	3.97	2.7	0.5872	1.48
14	0	0	0	0	2.961	3.955	0.4782	0.266	3.22	3.516	0.49579	0.568	3.97	2.7	0.5872	1.48
15	1	1	-1	-1	3.15	3.88	0.4819	0.364	3.42	3.45	0.4931	0.634	4.26	2.57	0.5432	1.75
16	0	0	0	0	2.961	3.955	0.4782	0.266	3.22	3.516	0.49579	0.568	3.97	2.7	0.5872	1.48
17	1	1	-1	1	2.74	3.92	0.47893	0.5	2.86	3.616	0.4927	0.173	3.37	2.916	0.518	0.963
18	-1	-1	-1	1	4.23	5.29	0.4732	0.568	4.26	4.93	0.47389	0.28	4.67	4.25	0.4847	0.836
19	-1	0	0	0	4.52	6.595	0.5317	0.219	4.7	6.18	0.5428	0.838	5.36	5.324	0.596	1.68
20	1	1	1	-1	3.87	4.73	0.729	1.404	4.15	4.31	0.8268	1.912	5.09	3.31	1.31	2.16
21	-1	1	-1	1	3.78	5.25	0.4792	0.483	3.84	4.896	0.49	0.221	4.3	4.16	0.5123	0.98
22	0	0	0	0	2.961	3.955	0.4782	0.266	3.22	3.516	0.49579	0.568	3.97	2.7	0.5872	1.48
23	0	-1	0	0	3.05	4.1	0.4736	0.311	3.289	3.66	0.49	0.52	4	2.895	0.5652	1.45
24	-1	-1	1	-1	6.24	6.85	0.6624	1.11	6.4	6.35	0.6957	1.57	7.1	5.359	1.2	1.75
25	1	-1	-1	1	3.14	4.82	0.479	0.456	3.35	4.32	0.4873	0.2	3.77	3.62	0.5127	0.775
26	1	-1	1	-1	3.46	4.51	0.715	1.41	3.72	4.1	0.9179	1.92	4.64	3.17	1.421	3.56
27	0	1	0	0	2.92	3.84	0.4764	0.265	3.17	3.379	0.49855	0.794	3.98	2.536	0.5654	1.72
28	0	0	0	0	2.961	3.955	0.4782	0.266	3.22	3.516	0.49579	0.568	3.97	2.7	0.5872	1.48
29	-1	1	-1	-1	4.48	5.134	0.4791	0.3	4.7	4.64	0.4904	0.69	5.46	3.7	0.5233	1.77
30	0	0	1	0	3.65	6.09	0.5457	0.6358	3.82	5.72	0.5623	0.953	4.47	4.87	0.7899	1.77
31	0	0	-1	0	3.52	4.487	0.4821	0.541	3.63	4.09	0.4902	0.191	4.16	3.348	0.5177	0.9955

Table 5

Range of variables and experiment Result for scenario (3) (three phase fault).

Exp.	K _{p1}	T _{i1}	K _{p2}	T _{i2}	X ₁ (MPUS ₁)	X ₂ (MPOS ₁)	X ₃ (T _{set1})	X ₄ (E _{ss1})	X ₅ (MPUS ₂)	X ₆ (MPOS ₂)	X ₇ (T _{set2})	X ₈ (E _{ss2})	X ₉ (MPUS ₃)	X ₃ (MPOS ₃)	X ₁₁ (T _{set3})	X ₁₂ (E _{ss3})
1	0	0	0	0	92.39	6.58	1.8859	0.3356	90.112	6.244	1.689	0.477	87.806	5.52	1.1669	0.771
2	-1	1	1	1	92.291	7.42	3	0.328	90	7.12	2.697	1,002,000	87.63	6.55	2.1166	0.585
3	1	1	1	1	92.31	7.53	2.786	0.4	90	7.2	2.68	0.9	87.66	6.634	2.186	0.55
4	-1	-1	-1	-1	92.54	7.37	5.135	1.7659	90.31	7.14	5.0034	1.498	88.049	6.833	5.678	0.987
5	1	0	0	0	92.389	6.738	1.8362	0.4929	90.101	6.39	1.686	0.351	87.79	5.68	1.16	0.807
6	-1	-1	1	1	92.294	7.28	6.987	1.987	89.987	6.97	6.154	0.799	87.654	6.37	5.9987	1.45
7	0	0	0	0	92.39	6.58	1.8859	0.3356	90.112	6.244	1.689	0.477	87.806	5.52	1.1669	0.771
8	1	-1	-1	-1	92.442	7.89	4.3	0.95	90.18	7.7177	4.134	1.09	87.88	7.38	4.5	0.899
9	1	-1	1	1	92.303	7.46	5.2	7.63	89.99	7.144	4.765	6.99	87.64	6.53	4.523	5.498
10	0	0	0	1	92.344	7.6	2.3	0.277	90.035	7.28	2.222	0.288	87.71	6.659	2.016	0.334
11	0	0	0	-1	92.677	3.76	0.553	0.4198	90.5	3.349	0.52862	0.3333	88.254	2.3	0.478	0.192
12	-1	1	1	-1	92.58	4.77	0.689	0.53	90.36	4.4	0.617	1.2	88.1	3.5	0.5283	923.4
13	0	0	0	0	92.39	6.58	1.8859	0.3356	90.112	6.244	1.689	0.477	87.806	5.52	1.1669	0.771
14	0	0	0	0	92.39	6.58	1.8859	0.3356	90.112	6.244	1.689	0.477	87.806	5.52	1.1669	0.771
15	1	1	-1	-1	92.54	7.8	3.202	0.274	90.32	7.61	3.2027	0.457	88.04	7.19	3.2028	0.355
16	0	0	0	0	92.39	6.58	1.8859	0.3356	90.112	6.244	1.689	0.477	87.806	5.52	1.1669	0.771
17	1	1	-1	1	92.3	8.17	6.2	4.56	89.995	7.91	6.078	8.76	87.665	7.49	5.987	6.76
18	-1	-1	-1	1	92.29	8.07	7.136	1.87	89.975	7.8	7.057	1.43	87.643	7.339	7.45	0.993
19	-1	0	0	0	92.38	6.9	2.108	0.559	90.075	6.57	2.041	0.506	87.758	5.9	1.3192	0.44
20	1	1	1	-1	92.57	4.6	0.622	0.51	90.35	4.21	0.5748	0.35	88.085	3.29	0.4865	0.375
21	-1	1	-1	1	92.3	7.88	7.176	8.033	89.98	7.61	6.854	7.995	87.64	7.239	6.3609	8.12
22	0	0	0	0	92.39	6.58	1.8859	0.3356	90.112	6.244	1.689	0.477	87.806	5.52	1.1669	0.771
23	0	-1	0	0	92.41	6.59	2.336	0.249	90.13	6.25	2.211	0.371	87.802	5.539	1.6637	0.332
24	-1	-1	1	-1	92.55	4.77	0.8887	0.51	90.33	4.41	0.8138	0.46	88.03	3.5	0.603	0.68
25	1	-1	-1	1	92.29	8.42	6.67	1.78	89.98	8.13	6.48	0.9876	87.66	7.69	6.706	2.56
26	1	-1	1	-1	92.58	4.94	0.7419	0.75	90.37	4.54	0.6919	0.87	88.081	3.61	0.529	1.2
27	0	1	0	0	92.428	6.7	1.5363	0.345	90.149	6.36	1.461	0.285	87.823	5.65	0.9861	0.471
28	0	0	0	0	92.39	6.58	1.8859	0.3356	90.112	6.244	1.689	0.477	87.806	5.52	1.1669	0.771
29	-1	1	-1	-1	92.55	7.7	6.87	5.6	90.33	7.539	6.567	5.1	88.05	7.08	6.126	6.1
30	0	0	1	0	92.38	7.19	2.361	4.52	90.074	6.86	2.213	0.347	87.742	6.18	1.638	0.525
31	0	0	-1	0	92.345	8.03	5	7.44	90.047	7.763	4.9	5.5	87.727	7.278	4.6	3.25

Table 6

The constant values for Scenario 1.

Cons.	scenario (1)								
	X_1	X_3	X_4	X_5	X_7	X_8	X_9	X_{11}	X_{12}
Z_1	14.783	0.14125	0.7454	-0.0094	0.1401	0.5251	15.896	0.16001	0.6290
Z_2	-5.335	0.01859	0.0037	-5.351	0.0694	0.0155	-5.377	0.02356	-0.0154
Z_3	-1.395	-0.00882	-0.0824	-1.559	-0.0611	-0.0448	-1.557	-0.00936	0.0864
Z_4	3.884	0.08121	0.0234	3.869	0.0388	0.0781	3.897	0.10921	0.0471
Z_5	-0.073	0.00490	0.2689	-0.067	0.0562	0.0605	-0.080	0.00491	-0.2858
Z_6	2.905	-0.0023	0.049	2.521	0.0168	-0.010	2.484	0.0038	0.1611
Z_7	-0.391	-0.0025	0.017	0.588	0.0166	0.083	0.622	-0.0008	0.0096
Z_8	-0.776	0.0112	-0.102	-1.190	0.0291	-0.020	-1.308	0.0220	0.0740
Z_9	0.479	0.0130	-0.107	0.092	0.0210	0.009	0.071	0.0065	0.0870
Z_{10}	0.648	0.01378	-0.0800	0.665	-0.0385	-0.0597	0.665	0.01883	-0.0805
Z_{11}	-2.394	0.01857	-0.1460	-2.392	-0.0391	-0.0777	-2.385	0.01529	0.0121
Z_{12}	-0.059	-0.00870	-0.0052	-0.066	0.0469	0.0039	-0.058	-0.01149	-0.0074
Z_{13}	0.341	0.00225	0.0801	0.341	0.0547	0.0623	0.353	-0.00617	0.0261
Z_{14}	-0.039	-0.00765	-0.0716	-0.046	-0.0649	-0.0986	-0.040	-0.00686	-0.0121
Z_{15}	-0.096	0.00763	-0.0094	-0.098	-0.0490	-0.0690	-0.090	0.00566	-0.0915

Table 7

The constant values for Scenario 2.

Cons.	scenario (2)											
	X ₁	X ₂	X ₃	X ₄	X ₅	X ₆	X ₇	X ₈	X ₉	X ₁₀	X ₁₁	X ₁₂
Z ₁	3.0002	4.118	0.47779	0.3454	3.2518	3.683	0.4932	0.6417	4.0041	2.866	0.5959	1.572
Z ₂	-1.0177	-1.0522	-0.01562	0.0467	-0.9684	-1.0263	-0.0025	0.0532	-0.9218	-0.9968	0.0167	0.2003
Z ₃	-0.0400	-0.0622	0.00610	0.0092	-0.0349	-0.0565	0.0042	0.0786	0.0114	-0.0849	-0.0036	-0.0312
Z ₄	0.3789	0.6742	0.07455	0.1698	0.3756	0.6617	0.0953	0.3778	0.4153	0.6084	0.2120	0.2013
Z ₅	-0.3488	0.1329	-0.03559	-0.2133	-0.4145	0.1586	-0.0559	-0.4084	-0.5831	0.2585	-0.1642	-0.4281
Z ₆	0.350	0.650	0.0197	-0.2250	0.320	0.655	0.0250	0.011	0.2772	0.662	-0.0130	-0.099
Z ₇	-0.061	-0.339	-0.0023	-0.1500	-0.059	-0.359	0.0041	-0.071	-0.0538	-0.344	-0.0407	-0.094
Z ₈	0.539	0.980	0.0366	0.1504	0.436	1.026	0.0361	-0.156	0.2712	1.050	0.0478	-0.296
Z ₉	0.218	-0.089	0.0252	0.5030	0.256	-0.124	0.0305	0.338	0.3372	-0.194	0.1473	0.341
Z ₁₀	0.0331	-0.0913	-0.00627	0.0259	0.0231	-0.0723	-0.0141	-0.0166	0.0246	-0.0682	-0.0039	-0.005
Z ₁₁	-0.4194	-0.4133	-0.01426	0.0350	-0.4119	-0.3773	-0.0016	0.0985	-0.3741	-0.3706	0.0121	0.270
Z ₁₂	-0.0294	-0.0483	-0.02016	-0.0674	-0.0169	-0.0460	-0.0320	-0.0140	-0.0116	-0.0499	-0.0375	-0.180
Z ₁₃	0.0481	0.0387	0.00406	0.0285	0.0519	0.0191	-0.0010	0.0173	0.0496	0.0174	-0.0137	-0.186
Z ₁₄	-0.1219	-0.1388	-0.00526	0.0404	-0.1306	-0.1306	0.0013	0.0175	-0.1354	-0.1244	0.0168	0.193
Z ₁₅	-0.1844	-0.0133	-0.03615	-0.2518	-0.1881	-0.0321	-0.0576	-0.2178	-0.2066	-0.0268	-0.1462	-0.024

Table 8
The constant values for Scenario 3.

Cons.	scenario (3)											
	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8	X_9	X_{10}	X_{11}	X_{12}
Z_1	92.4010	6.581	1.775	0.822	90.1204	6.239	1.630	0.492	87.8112	5.506	1.102	−6.8
Z_2	−0.00283	0.077	−0.468	−0.213	−0.00339	0.072	−0.417	0.087	−0.00294	0.066	−0.383	−51.3
Z_3	0.00944	−0.012	−0.406	0.172	0.01289	−0.008	−0.365	0.598	0.01411	−0.009	−0.537	51.8
Z_4	0.01450	−0.854	−1.579	−0.839	0.01911	−0.909	−1.615	−1.150	0.01489	−1.075	−1.778	50.2
Z_5	−0.12817	0.902	1.359	0.864	−0.17267	0.903	1.270	0.945	−0.20372	0.990	1.178	−50.4
Z_6	−0.0293	0.236	0.327	−0.864	−0.0423	0.246	0.302	−0.081	−0.0433	0.299	0.213	16.3
Z_7	0.0052	0.062	0.291	−1.093	0.0092	0.071	0.274	−0.181	−0.0048	0.104	0.298	16.0
Z_8	−0.0513	1.027	2.035	4.590	−0.0698	1.078	1.995	2.414	−0.0828	1.238	2.093	17.5
Z_9	0.0967	−0.903	−0.219	−1.042	0.1372	−0.919	−0.186	−0.199	0.1647	−1.011	0.221	15.9
Z_{10}	0.00363	−0.056	−0.106	−0.858	0.00481	−0.059	−0.078	−0.612	0.00906	−0.058	0.013	−58.5
Z_{11}	0.00975	−0.061	0.233	0.977	0.00956	−0.068	0.251	0.698	0.01181	−0.070	0.231	−57.1
Z_{12}	0.00725	0.019	0.080	0.505	0.00831	0.018	0.102	0.793	0.01256	0.018	0.106	58.3
Z_{13}	−0.00650	0.004	−0.433	−1.326	−0.00919	−0.001	−0.368	−1.485	−0.00581	0.013	−0.230	56.3
Z_{14}	−0.00650	−0.008	−0.447	−0.180	−0.00794	−0.010	−0.404	0.279	−0.01481	0.015	−0.441	−57.6
Z_{15}	−0.01188	0.552	0.460	0.024	−0.01394	0.589	0.377	−0.313	−0.01881	0.682	0.355	−58.0

Table 9
The input weights of the multi objective function.

Weight (Wt)	Value
W_{t1} (For DEG1 MPUS equation)	0.2
W_{t2} (For DEG1 MPOS equation)	0.2
W_{t3} (For DEG1 T_{set} equation)	0.075
W_{t4} (For DEG1 E_{ss} equation)	0.03
W_{t5} (For DEG2 MPUS equation)	0.125
W_{t6} (For DEG2 MPOS equation)	0.125
W_{t7} (For DEG2 T_{set} equation)	0.04
W_{t8} (For DEG2 E_{ss} equation)	0.02
W_{t9} (For DEG3 MPOS equation)	0.075
W_{t10} (For DEG3 MPUS equation)	0.075
W_{t11} (For DEG3 T_{set} equation)	0.025
W_{t12} (For DEG3 E_{ss} equation)	0.01

4.2. The SFO algorithm

The growth of soft computational capabilities is the key motivation for using the SFO to find solution for the problems to be optimized. The SFO is a population-based heuristic one which is inspired by the nature. Its main concept is to simulate the sunflowers' orientation to get the radiation of the sun [59]. The sequence of a sunflower is daily periodic. They begin the day with a sun-tracking movement. At the evening, they go through the reverse direction. Each sunflower is assumed to create only one pollen gamete. The sunflowers directions towards the sun presented in Eq. (1):

$$\vec{s}_i = \frac{X^* - X_i}{\|X^* - X_i\|}, \quad i = 1, 2, \dots, n_p. \quad (1)$$

Eq. (2) shows the sunflowers' step towards the direction s :

$$d_i = \lambda \times P_i(X_i + X_{i-1}) \times \|X_i + X_{i-1}\|, \quad (2)$$

where λ is a constant describing an "inertial" movement of the sunflowers, $P_i(\|X_i + X_{i-1}\|)$ is the pollination probability. The nearest population to the sun makes smaller movements to seek a local refinement. Meanwhile, the more distant individuals' movements are normal. The restriction on these steps is introduced in Eq. (3):

$$d_{max} = \frac{\|X_{max} - X_{min}\|}{2 \times N_{pop}}, \quad (3)$$

where X_{max} and X_{min} are the upper and lower limits, and N_{pop} is the total populations number of the plants. The new plant is given by this equation:

$$\vec{X}_{i+1} = \vec{X}_i + d_i \times \vec{s}_i. \quad (4)$$

The SFO algorithm steps are simply:

- 1) Production of the population which is random.
- 2) The population with the top evaluation is converted to the sun.
- 3) These populations modify their orientation towards the sun.

For more explanation, a flowchart for the algorithm is shown in Fig. 3.

5. Design procedures

5.1. Selection of variables and levels

In this system we have 6 PI controllers which are divided into 2 PI controllers in each DG where:

$PI_{1,1}$ and $PI_{1,2}$ are the PI controllers in DG₁
 $PI_{2,1}$ and $PI_{2,2}$ are the PI controllers in DG₂
 $PI_{3,1}$ and $PI_{3,2}$ are the PI controllers in DG₃

The proportional gain (PK) and integral time constants (IT) are the design values of the PI controllers in this system where:

K_1 : proportional gain of DG₁ ($K_1 = PK_{1,1}$),
 K_2 : integral time constant of DG₁ ($K_2 = IT_{1,1}$),
 K_3 : proportional gain of DG₁ ($K_3 = PK_{1,2}$),
 K_4 : integral time constant of DG₁ ($K_4 = IT_{1,2}$),
 K_5 : proportional gain of DG₂ ($K_5 = PK_{2,1}$),
 K_6 : integral time constant of DG₂ ($K_6 = IT_{2,1}$),
 K_7 : proportional gain of DG₂ ($K_7 = PK_{2,2}$),

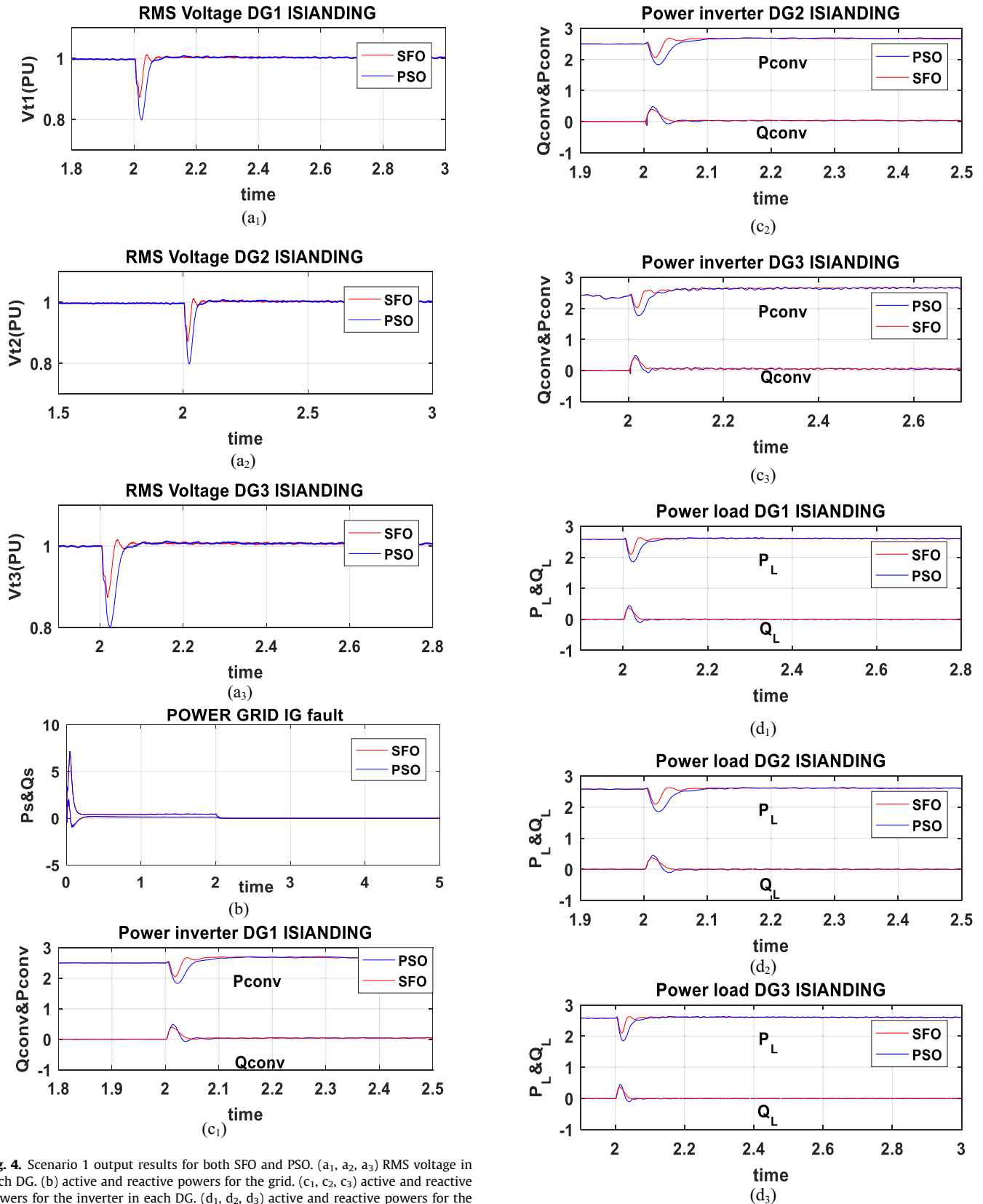


Fig. 4. Scenario 1 output results for both SFO and PSO. (a₁, a₂, a₃) RMS voltage in each DG. (b) active and reactive powers for the grid. (c₁, c₂, c₃) active and reactive powers for the inverter in each DG. (d₁, d₂, d₃) active and reactive powers for the load in each DG.

Fig. 4 (continued)

K_8 : integral time constant of DG₂ ($K_8 = IT_{2,2}$),
 K_9 : proportional gain of DG₃ ($K_9 = PK_{3,1}$),
 K_{10} : integral time constant of DG₃ ($K_{10} = IT_{3,1}$),
 K_{11} : proportional gain of DG₃ ($K_{11} = PK_{3,2}$),
 K_{12} : integral time constant of DG₃ ($K_{12} = IT_{3,2}$).

Three design levels are used for the controllers' variables in this system: minimum value (Level 1), average value (Level 2), and maximum value (Level 3). These levels are presented in Table 2.

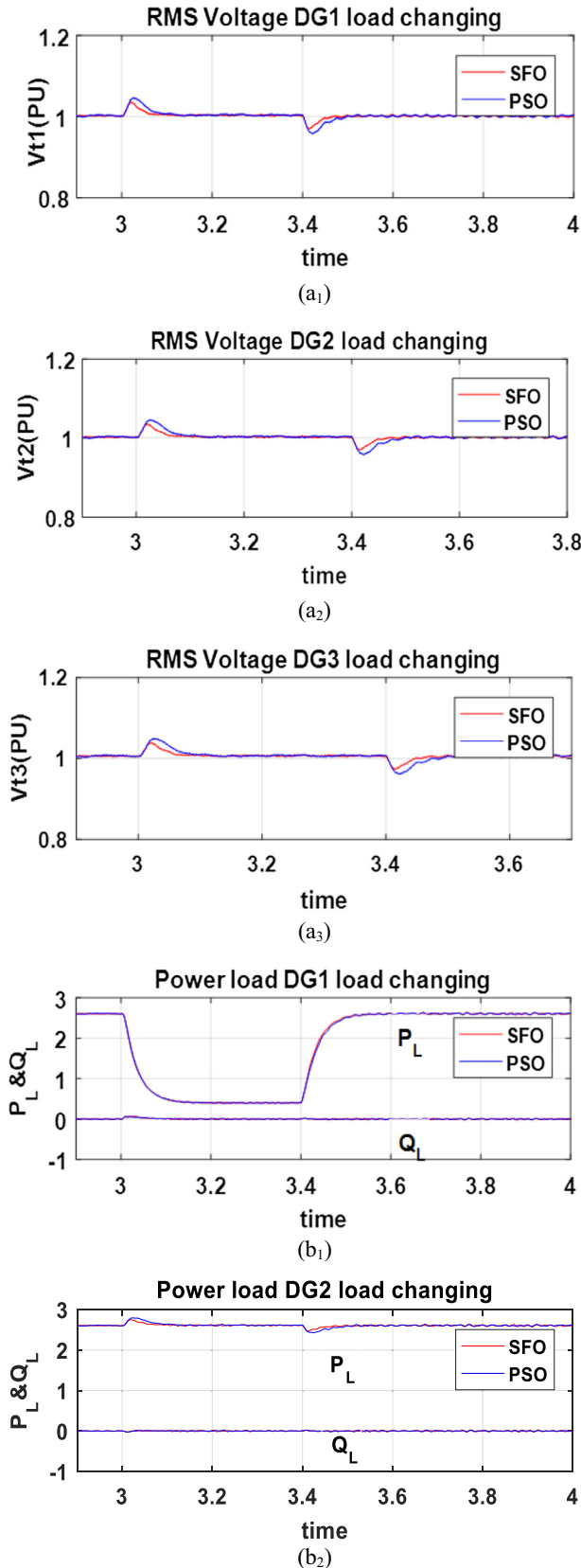


Fig. 5. Scenario 2 output results for both SFO and PSO. (a₁, a₂, a₃) RMS voltage in each DG. (b₁, b₂, b₃) active and reactive powers for the load in each DG.

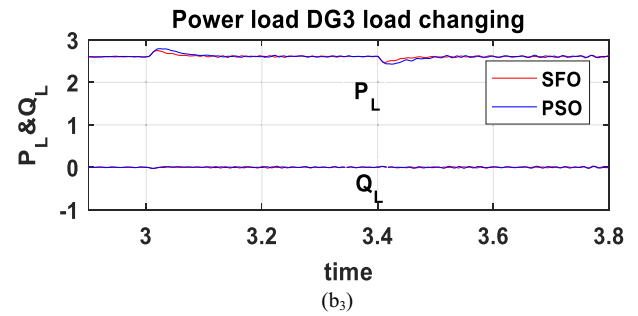


Fig. 5 (continued)

5.2. PSCAD/EMTDC program calculation

The calculations results of PSCAD program for each scenario are executed. In scenario 1, the system is converted from the grid-connected to the stand-alone mode, while in scenario 2, the load is changing during the stand-alone mode. And in scenario 3, Symmetrical Fault is considered. The values of Tset, Ess, MPUS, Tset and MPOS are shown in Table 3, 4, and 5.

5.3. Response surface empirical targets determination

The minimization of the MPOS (X_1), MPUS (X_2), T_{set} (X_3), and E_{ss} (X_4) for the presented scenarios defines the multi-objective function. Eq. (5) shows the second-order polynomial function extracted from RSM model for scenario 1.

$$X_i = Z_1 + Z_2K_1 + Z_3K_2 + Z_4K_3 + Z_5K_4 + Z_6K_1^2 + Z_7K_2^2 + Z_8K_3^2 + Z_9K_4^2 + Z_{10}K_1K_2 + Z_{11}K_1K_3 + Z_{12}K_1K_4 + Z_{13}K_2K_3 + Z_{14}K_2K_4 + Z_{15}K_3K_4 \quad (5)$$

where

$$i = 1, 2, 3, 4$$

$Z_1, Z_2, \dots, Z_{15}$ = the RSM model constants obtained for each scenario which are presented in Table 6–8.

5.4. Optimization using SFO

The multi-objective function Eq. (5) based on the weighting method [60] is used as an input to the SFO code stated in [61]. The settings of the SFO are 30 flowers and 600 iterations. The weights used in the multi objective function are stated in Table 9, where (W_i) is the weight selected. Their summation equals 1.

6. Simulation results and discussion

The simulation of the system is done by PSCAD/EMTDC. Three scenarios have been tested in this paper. Scenario 1 (STAND ALONE MODE) starts at $t = 2$ s. In scenario 2 (variable load mode), R_{12} is changed from 150Ω to 300Ω at $t = 3$ s and then back again to its original value at $t = 3.4$ s. In the scenario 3, a symmetrical fault is applied at $t = 4$ s and is cleared at $t = 4.1$ s. The scenarios simulation results are introduced in Figs. 4–6.

PI (1) and PI (2) controllers' parameters are achieved using PSO and SFO techniques. The SFO and PSO codes are constructed by the MATLAB program. The results are represented in Table 10. They verify the flexibility, justification, and applicability of the presented SFO algorithm versus the particle swarm optimization (PSO).

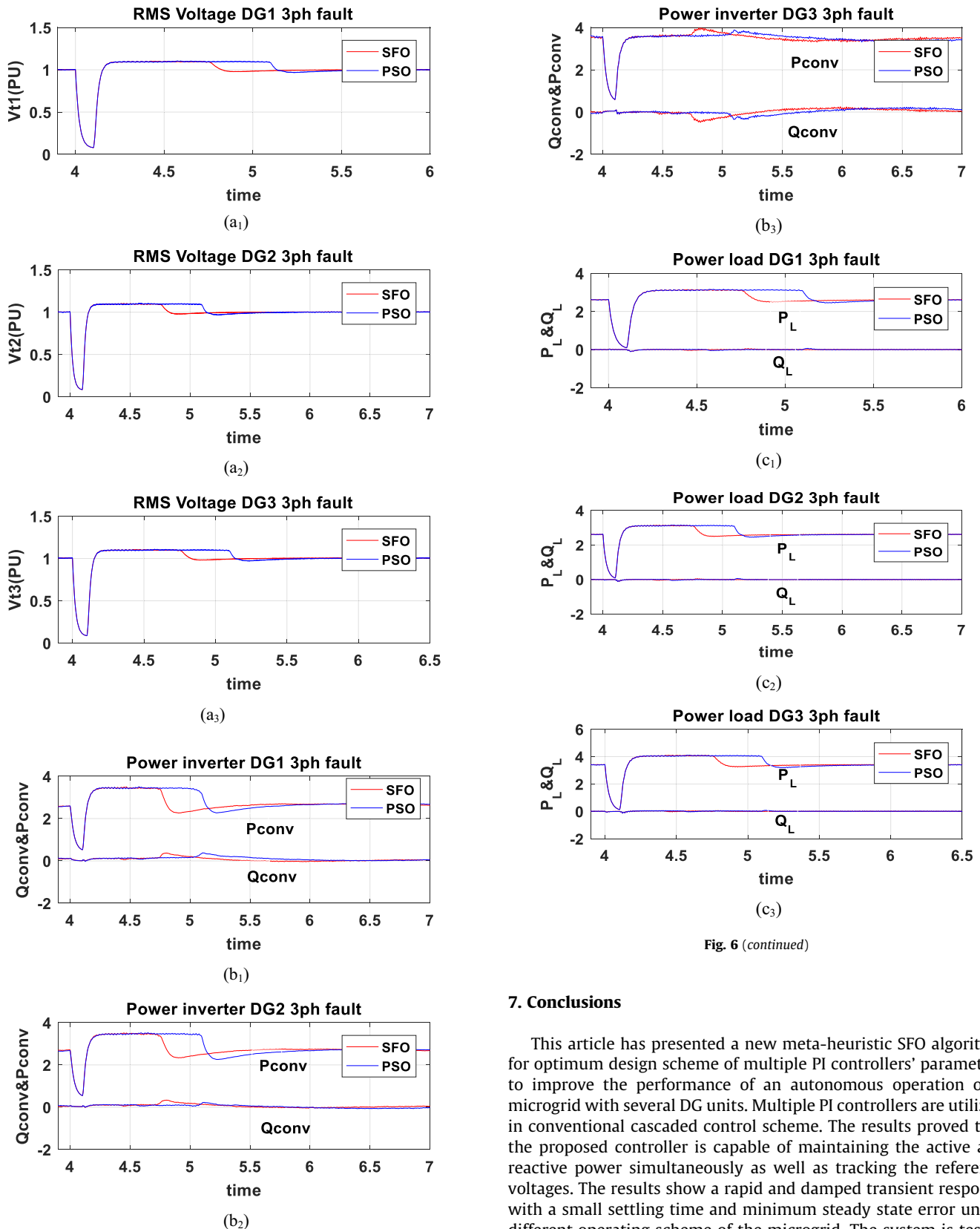


Fig. 6 (continued)

Fig. 6. Scenario 3 output results for both SFO and PSO. (a₁, a₂, a₃) RMS voltage in each DG. (b₁, b₂, b₃) active and reactive powers for the inverter in each DG. (c₁, c₂, c₃) active and reactive powers for the load in each DG.

7. Conclusions

This article has presented a new meta-heuristic SFO algorithm for optimum design scheme of multiple PI controllers' parameters to improve the performance of an autonomous operation of a microgrid with several DG units. Multiple PI controllers are utilized in conventional cascaded control scheme. The results proved that the proposed controller is capable of maintaining the active and reactive power simultaneously as well as tracking the reference voltages. The results show a rapid and damped transient response with a small settling time and minimum steady state error under different operating scheme of the microgrid. The system is tested under three different operating states which are: 1) system conversion from the grid-connected to the stand-alone mode, 2) changing

Table 10

A Comparative Evaluation For The Results Obtained From SFO And PSO Techniques For The Three Proposed Scenarios.

Point Of comparison	SFO						PSO					
	DG1		DG2		DG3		DG1		DG2		DG3	
Scenario 1												
Optimum size	Y ₁	6.4	Y ₅	5.98	Y ₉	5.534	Y ₁	2.146	Y ₅	1.569	Y ₉	1.067
	Y ₂	0.0052	Y ₆	0.00408	Y ₁₀	0.00311	Y ₂	0.0056	Y ₆	0.0042	Y ₁₀	0.00311
	Y ₃	2.95	Y ₇	2.508	Y ₁₁	2.099	Y ₃	1.678	Y ₇	1.234	Y ₁₁	0.992
	Y ₄	0.347	Y ₈	0.2987	Y ₁₂	0.24786	Y ₄	0.338	Y ₈	0.3056	Y ₁₂	0.2589
MPUS	12.905%		12.86%		12.6%		20.26%		20.1%		19.975%	
T _{set}	0.033 Sec		0.0326 sec		0.0322 sec		0.054 Sec		0.0535 sec		0.0531	
E _{ss}	0.35%		0.36%		0.32%		0.4%		0.405%		0.402%	
Scenario 2												
Optimum size	Y ₁	6.4687	Y ₅	6.0241	Y ₉	5.497	Y ₁	1.921	Y ₅	1.4023	Y ₉	0.8997
	Y ₂	0.01249	Y ₆	0.0093	Y ₁₀	0.0066	Y ₂	0.0115	Y ₆	0.0101	Y ₁₀	0.0653
	Y ₃	2.279	Y ₇	1.861	Y ₁₁	1.578	Y ₃	2.311	Y ₇	1.7981	Y ₁₁	1.4877
	Y ₄	0.237	Y ₈	0.2072	Y ₁₂	0.175	Y ₄	0.2312	Y ₈	0.1998	Y ₁₂	0.1623
MPUS	3.2%		3.12%		2.745%		4.3%		4.275%		3.909%	
MPOS	3.97%		3.61%		3.6%		5%		4.64%		4.7%	
T _{set}	0.43 sec		0.4289 sec		0.424 sec		0.45 sec		0.4455 sec		0.443 sec	
E _{ss}	0.44%		0.665%		1.032%		0.48%		0.69%		1.07%	
Scenario 3												
Optimum size	Y ₁	6.134	Y ₅	6.2	Y ₉	6.1	Y ₁	2.108	Y ₅	2.176	Y ₉	2.2
	Y ₂	0.004	Y ₆	0.0041	Y ₁₀	0.004	Y ₂	0.00613	Y ₆	0.00599	Y ₁₀	0.006
	Y ₃	2.498	Y ₇	2.51	Y ₁₁	2.47	Y ₃	2.57	Y ₇	2.55	Y ₁₁	2.5
	Y ₄	0.1211	Y ₈	0.119	Y ₁₂	0.12	Y ₄	0.1	Y ₈	0.0987	Y ₁₂	0.0105
MPUS	91.638%		92.135%		93.106%		93.108%		92.14%		91.64%	
MPOS	10.705%		10.3505%		10.205%		10.9564%		10.604%		10.59%	
T _{set}	0.9654 sec		0.9654 sec		0.8069 sec		1.3901 sec		1.3901 sec		1.3842 sec	
E _{ss}	0.46%		0.495%		0.842%		0.55%		0.63%		1.02%	

the load during the stand-alone mode, and 3) Applying symmetrical fault during the stand-alone mode. The results verify the flexibility, justification, and applicability of the presented SFO algorithm versus the particle swarm optimization (PSO). The validity of the introduced system is proved based on simulation results. It achieved by PSCAD/EMTDC under several operating scenarios of the microgrid. The MINITAB software is utilized to get the constraints and the objective function. The SFO and PSO codes are constructed by the MATLAB software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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